SmartSDR™ for Windows v2.12.1 Release Notes

March 12, 2024

SmartSDR™ v2.12.1 is a hot fix release for all FLEX-6000-series (FLEX-6400™, FLEX-6400M™, FLEX-6600™, FLEX-6600M™, FLEX-6300™, FLEX-6500™, and FLEX-6700™) software-defined radios and the Maestro™. This version supersedes all previous versions of SmartSDR v2.x.

This document covers changes to client software (SmartSDR for Windows, SmartSDR for the "M" Model FLEX-6000s, and Maestro). Changes to clients or utilities produced by 3rd parties are not covered. Please consult the SmartSDR Software User Guide V2 for details on using features and an optimal operating experience.

Before upgrading, please read all sections marked as IMPORTANT for upgrade prerequisites and other important information. Failure to follow these instructions may create problems requiring FlexRadio support or service staff interventions.

What's New in SmartSDR v2.12.1?

Recent versions of SmartSDR v2 address several critical software defects identified in previous versions of SmartSDR.

VERY IMPORTANT: After upgrading to SmartSDR v2.12.1, downgrading to a previous version of SmartSDR v2 will result in the reoccurrence of critical issues resolved in this software release.

Notice: To better support current and future hardware products, **s**tarting with SmartSDR v2.12.0, the file size of the SmartSDR for Windows installer software package has increased in size. We have made improvements to the file distribution infrastructure for faster downloads globally. Please exercise patience if it takes a little longer to download the SmartSDR for Windows installer software.

New Features in SmartSDR v2.12.1

No new features are included in the SmartSDR v2.12.1 hot fix release as the focus was on addressing a critical Maestro "C" operational issue.

Bug Fix Highlights in SmartSDR v2.12.1

Please refer to the SmartSDR v2 Changelog for a complete listing of changes.

[SMART-10129] Updated Maestro "C' Windows operating system's user credentials to prevent future Maestro boot problems.

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Important Release Details

What to do if the FLEX-6000 fails to boot after upgrading to SmartSDR v2.12.1: If the FLEX-6000 fails to boot after performing a firmware upgrade, please open a HelpDesk support ticket to resolve the issue.

Backup your profiles before upgrading: Moving back and forth between different versions of SmartSDR may result in data loss unless you have a backup (export) of your Profiles. It is recommended to export all profiles *before upgrading* to v2.12.1.

Exporting profile data is covered in detail below in the *Best Practices for Installing / Uninstalling SmartSDR* v2 section. Maintaining a good set of profile exports as backups is *always* a best practice.

Upgrading FLEX-6400M and FLEX-6600M Front Panel: If you are upgrading a FLEX-6400M or a FLEX-6600M using SmartSDR for Windows, the upgrade will not be applied to the software for the Front Panel of the "M" Model. Use the information below to upgrade the front panel.

- ***The radio must first be connected to your network and can access the Internet to download the necessary software to update the front panel.***
- After the radio firmware updates from your PC, shut down SmartSDR for Windows and connect to your radio using the "M" Model Front Panel. The Version Select screen should display SmartSDR v2.12.1 showing a download icon. Selecting that version will start the final portion of the upgrade process.

For a detailed description of the SmartSDR for Windows software upgrade process, please refer to the SmartSDR Software Users Guide v2

Always Perform a Factory Reset of your Radio when Downgrading the Radio Firmware: In general, downgrading to a previous version of SmartSDR is not recommended. The database in the FLEX-6000 is NOT backward compatible. This means if you downgrade the firmware in your radio, you *must perform a Factory Reset* to ensure the internal database is consistent with the version of SmartSDR firmware that is running on the radio.

The procedure for performing a Factory Reset is described in the **Best Practices for Installing a New Version of SmartSDR for Windows section** below.

In addition to performing a Factory Reset on the radio, you mustn't import a database using a profile export that is greater than the version you are using with the FLEX-6000. In later versions of SmartSDR, the profile export file name contains the version of SmartSDR that was running when the export was saved to your PC for easy version identification.

Upgrading the radio firmware over a SmartLink connection is not supported. Since Internet connections can be inherently unreliable and it is paramount that the firmware upgrade process proceeds uninterrupted to prevent the FLEX-6000 from becoming inoperable, upgrading the radio firmware is only supported using a LAN or network-connected SmartSDR client (SmartSDR for Windows, the M model front panel and a Maestro).

Best Practices for Installing / Uninstalling SmartSDR

Best Practices for Installing a New Version of SmartSDR for Windows

***Always* back up your Global, TX (transmit), and Microphone Profiles.** Please refer to the *SmartSDR for Windows Software User Guide* for detailed instructions related to exporting your profiles to a file on your PC.

When you upgrade to a newer version of SmartSDR for Windows, a database conversion program runs automatically to convert older profile schemas to updated ones.

When downgrading versions, SmartSDR is unable to restore the updated profile schema version, which may cause profile corruption. To recover your previously created profiles after a software version downgrade, import the profile data created using the downgraded version of SmartSDR for Windows.

Ensure your Windows operating system is up to date before installing SmartSDR for Windows. The proper operation of SmartSDR for Windows and its associated drivers relies on having an up-to-date and supported Windows operating system including the root security certificates. It is strongly recommended that you run Windows Update and install all mandatory and optional updates before installing SmartSDR for Windows.

Managing SmartSDR installed on other devices. If using Maestro or other PCs running SmartSDR for Windows, update all devices at the same time to ensure a consistent operating experience and to avoid radio firmware upgrade/downgrade delays.

Power cycle the FLEX-6000 radios before installing a new version of SmartSDR for Windows. To ensure a seamless upgrade of the FLEX-6000, it is recommended that you power cycle your radio *before* installing the SmartSDR for Windows software on your PC and updating the radio firmware.

"Cold Boot" your FLEX-6000 after upgrading the radio firmware. Using the power button, shut down the radio, then disconnect it from DC power for 30 seconds. This procedure "*cold boots*" the radio and helps ensure proper operation. Once DC power is restored to the radio, wait an additional 2 minutes to ensure that all internal processors have booted up completely before pressing the power button to load the SmartSDR firmware and boot your radio.

What to do if the firmware updates do not finish after 10 minutes? If ten (10) minutes have elapsed without the radio restarting and completing the upgrade, power off the radio by pressing the power button. The radio should restart, and the firmware update should be completed normally.

Note: It would be rare that a single press on the power button does not shut down the radio. However, if this occurs, remove the DC power cable from the radio to force a hard reset. Wait about 1 minute, restore DC power, wait 2 minutes, and depress the power button. The radio will restart and the firmware update should complete successfully.

How to Factory Reset a FLEX-6000

There may be situations where a factory reset of the FLEX-6000 is needed to ensure that the radio is working properly. If instructed to Factory Reset the FLEX-6000, use the procedure below.

Perform a Comprehensive Cold Boot and Factory Reset of your radio

NOTE: Before starting, shut down all programs that may interface with the radio.

Please pay attention to the "wait times" that are part of the steps listed below. These are important to ensure the radio resets properly.

NOTE: If you have a cable connected to the REM ON connector, please remove it before running the reset procedure. Reconnect it once the procedure has been completed.

- Power off the radio by pressing and releasing the power button. Allow it to completely power down before continuing by waiting for the power LED to turn off (or amber if a GPSDO is installed). If pressing and releasing the power button does not shut down the radio, **press and hold** the power button until the radio shuts down. If this does not work, turn off the DC power supply to shut down the radio.
- Once the radio is powered off, **wait for 2 minutes** to allow for all processors to shut down properly.
- Remove the power cable from the radio for **at least 30 seconds** and then reconnect it. It is important to remove the power cable from the radio and not just turn off the power supply.
- After reconnecting the DC power cable, turn the DC power ON and **wait for 2 minutes** to allow the internal PSoC processor to boot up completely before continuing.

For FLEX-6700 and FLEX-6500:

- Press and hold the **OK** button *while pressing and releasing* the **Power button**.
- Release the **OK** button once the power LED turns white and allow the radio to continue booting normally.

For FLEX-6300, FLEX-6400(M), and FLEX-6600(M):

- *Press and hold* the **Power button** for approximately 5 seconds until the Power button LED turns **white**. *After the power button turns white, release the **Power button*** and allow the radio to continue booting normally. Depending on different network factors, it may take up to 2 minutes for the radio to finish booting so please be patient and allow it to boot up completely.

Best Practices for installing a new version of SmartSDR for an "M" Model FLEX-6000 or Maestro

The following best practices are applicable only when installing a new version of SmartSDR on a Maestro.

Both the Maestro and an M Model FLEX-6000 must have network access that allows connectivity to the Internet to download the new SmartSDR software.

Ensure your Maestro has a reliable power source. Make certain the supplied AC adapter is used to power the unit. This prevents the Maestro from losing power during an update.

Ensure your Maestro has a reliable network connection. The Maestro and radio firmware are upgraded entirely through the network connection. When upgrading a Maestro, a wired Ethernet connection is recommended; wired Ethernet connections are faster and more reliable than Wi-Fi.

Uninstalling Previous Versions of SmartSDR for Windows – Is It Necessary?

In general, the answer is no, but there are considerations when more than one version of SmartSDR is maintained on your system:

Currently, every version of SmartSDR for Windows is installed in a "side by side" manner allowing the use of previous versions and supporting easy version switching of both software and radio firmware as long as there are no software dependencies that prevent backward compatibility. This method of installing places unique versions of software on your PC rather than removing or upgrading previous versions.

SmartSDR for Windows Shared Components:

The SmartSDR ecosystem employs shared components used by multiple versions of the software. When previous versions of SmartSDR for Windows are uninstalled it may result in the removal of one or more of these shared components, which may make newer versions inoperable.

Therefore, if you desire to uninstall an older version of SmartSDR, we highly recommend you uninstall older versions before installing a new version of SmartSDR for Windows.

Removing the DAX and FlexVSP drivers is not recommended unless explicitly instructed to do so in the Release Notes or by FlexRadio Support.

Known Issues

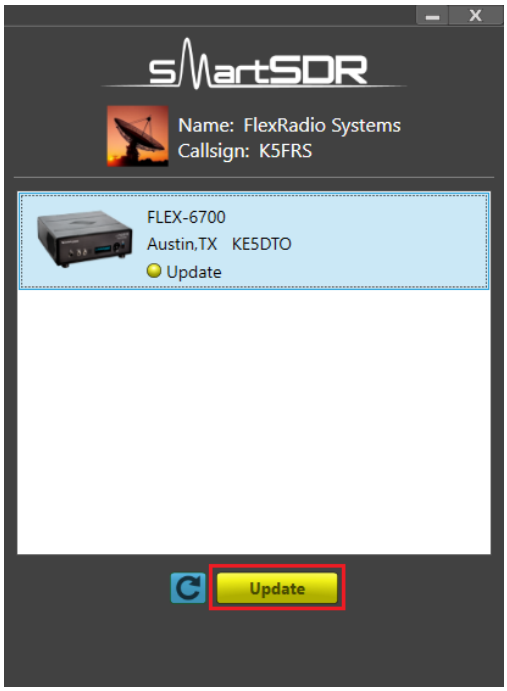
- **Reverting to previous versions of SmartSDR v2 is not recommended. However, if you revert from any version of SmartSDR v2 to a previous version you must perform a factory reset of your radio.** When reverting to a previous version of SmartSDR there can be database incompatibilities that may result in operational anomalies. It is recommended that a factory reset be performed after downgrading to ensure the database schema is 100% compatible with the version being used.

  Once the radio has been factory reset, **DO NOT** import any profiles created using a version of SmartSDR greater than the one installed on the FLEX-6000.

- During alpha testing, after the initial installation of the updated DAX drivers provided in SmartSDR v2.6.0, a handful of users reported a Windows "stop error" (colloquially known as the Blue Screen of Death or BSoD). In most cases, the error occurred once and after a subsequent PC reboot, the error did not occur again or impact proper DAX operations. We are working to identify the cause and resolve the behavior.

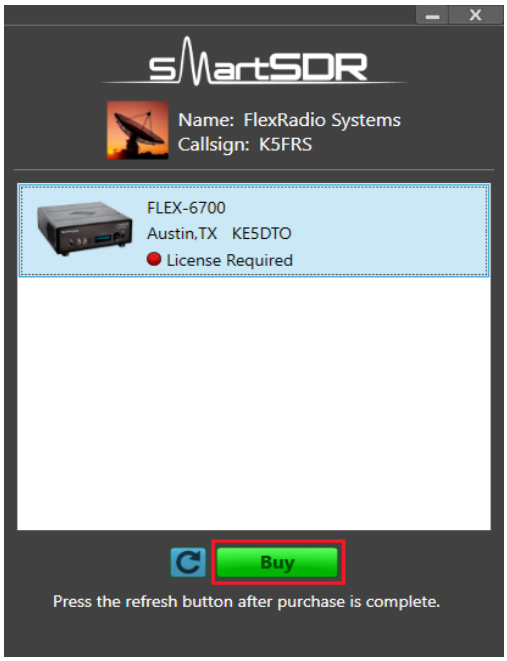
Purchasing a SmartSDR Software License for your FLEX-6000 using SmartSDR for Windows

Obtaining a SmartSDR software license is easy and convenient. Download the latest software installer for SmartSDR and run it to install the software on your PC. Using the new software, update the radio with the new firmware. Start SmartSDR then select the radio in the radio chooser window and press the yellow Update button.



Once the update completes, if the radio you are upgrading requires a license, the radio status will show "License Required" and you will be presented with a Buy button that will launch a web browser, which will take you through the steps to purchase the license.

Note: New radios purchased include a SmartSDR license for the current major version and will not see the "License Required" indicator.

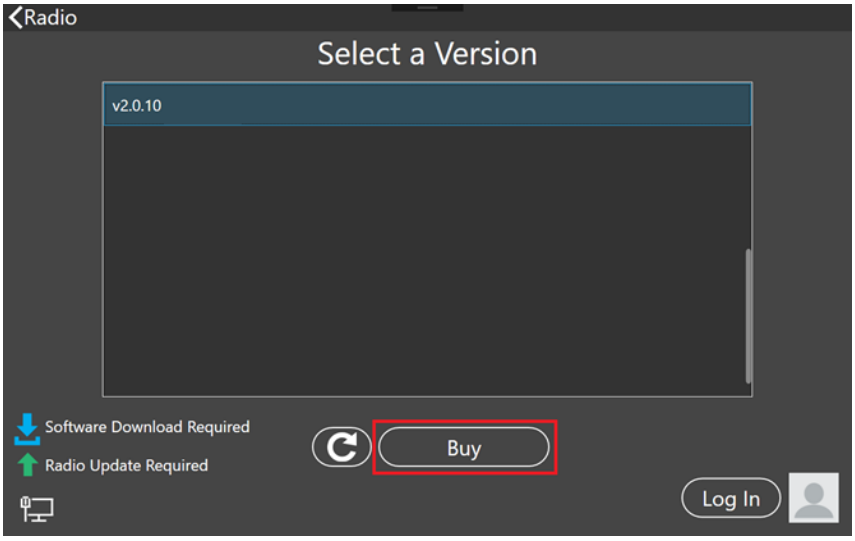


Once you have purchased a SmartSDR license for your radio, click the blue circular refresh button in the radio chooser window to update the license information for the radio. It may take up to a minute after the purchase completes before the radio acquires its license and cycling the power on the radio may be required.

Note: The radio (not just the computer) will need internet access to obtain the newly acquired license. Please ensure that the radio is plugged into a network that gives internet access.

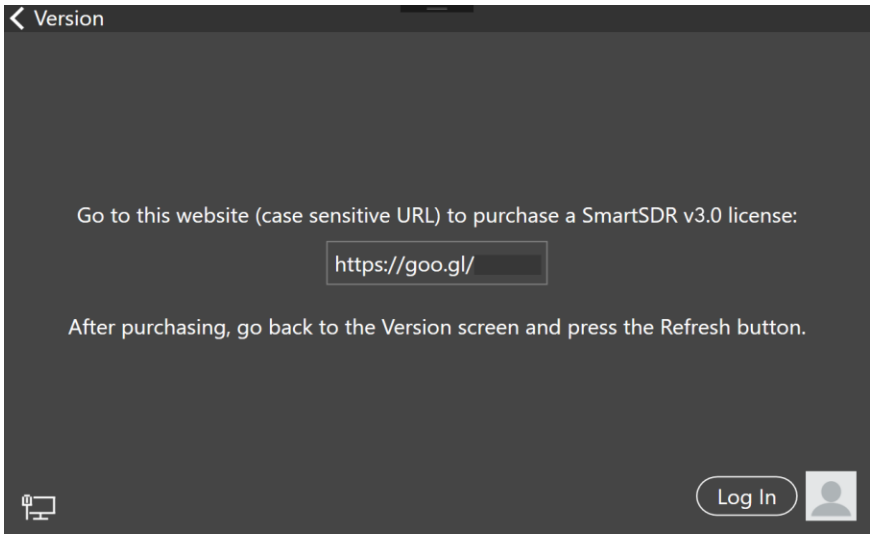
Purchasing a SmartSDR Software License for your FLEX-6000 using an "M" Model FLEX-6000 or a Maestro

Obtaining a SmartSDR software license using an M model FLEX-6000 or a Maestro is easy and convenient. After downloading and installing the latest SmartSDR software version. At the Select a Version screen, choose the latest version of the software just installed. If you need to purchase a license to run the software, the Buy button will appear as shown below.

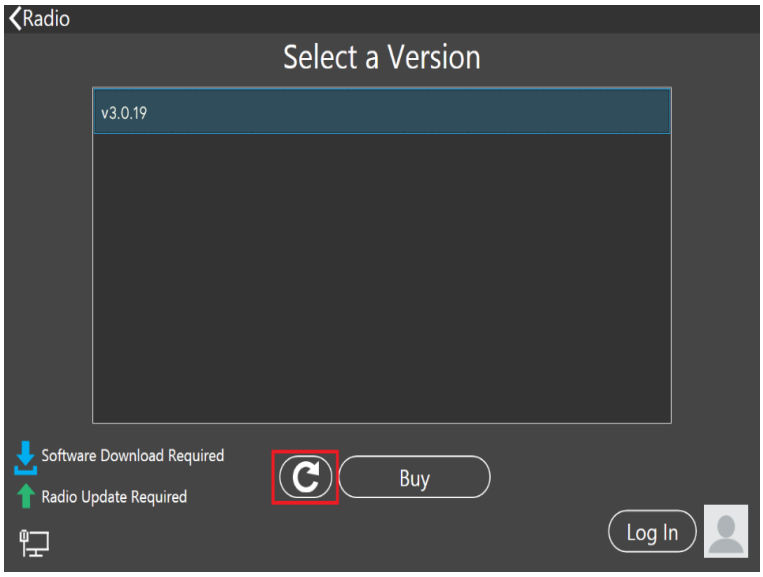


If you see a Run button, tap on the Radio link at the top left of the screen to return to the Select Radio to upgrade the radio's firmware.

Tapping the Buy button will generate and show a shortened URL that must be plugged into a browser manually. Note that the URL is case-sensitive and unique for each radio.



When the license purchase is complete, tap the Version link at the top left corner of the screen to go back to the Select a Version screen. At the Select a Version screen, tap the refresh button (circular icon) to update the license information for the radio. It may take up to a minute after the purchase is completed before Maestro finds the new license.



After the license is updated, the refresh button will disappear and the Buy button will change to a Run button. Tap the Run button to connect SmartSDR for an M Model FLEX-6000 or Maestro to your FLEX-6000.

SmartSDR Documentation Available from www.flexradio.com

The following documentation and how-to guides for SmartSDR v2 are available as a convenient download from the FlexRadio website.

SmartSDR for Windows Software User Guide

FLEX-6400M and FLEX-6600M User Guide

SmartSDR CAT User Guide

FLEX-6000 Signature Series Hardware Reference Manual

FLEX-6400 and FLEX-6600 Hardware Reference Manual

FLEX-6000 Signature Series Quick Start Guide

SmartLink for SmartSDR Quick Start Guide

USB Cable Interface Guide

Obtaining Technical Support

If you encounter any issues installing or operating SmartSDR for Windows with your FlexRadio Signature Series software defined radio, please use our online Community to search for information about SmartSDR and the FLEX-6000. If you need assistance using the Community, please refer to the Community Message Board for additional information.

If you are unable to find an existing answer to your issue via the Community, please contact FlexRadio Technical Support by opening a HelpDesk support ticket online.

For details on how to submit a HelpDesk support ticket, please refer to the HelpDesk article How to Submit a Request for Technical Support.

Hours of Operation: Our Technical Support engineers are available Monday-Friday from 7:00 AM-4:00 PM Central Time. If you open a HelpDesk ticket after business hours, on a holiday, or weekend, we will respond to your request for assistance during regular business hours in the order your HelpDesk ticket was received.

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